

ELEC3609/9609 · ASSIGNMENT 1

StudyCrew System & Database Design

Functional requirements, physical data model and complete interface wireframes

A design for visible work, earlier intervention and fairer team conversations.

Team	ELEC3609-Fri13-16-G02
Version	1.0 · 20 August 2026
Submission	Final system design report
Product	StudyCrew

Document control

Version	Date	Status	Scope
0.1	20 Aug 2026	Draft	Concept and requirement baseline
0.7	20 Aug 2026	Reviewed	ERD, wireframes and traceability
1.0	20 Aug 2026	Submission candidate	Integrated report

Contents

Section	Evidence delivered
1. Product definition	Problem, users, value and design boundary
2. Functional requirements	18 measurable IEEE 830-style requirements
3. Data model	13-table 3NF model, Crow's Foot ERDs and constraints
4. Wireframes	11 generic views covering every requirement
5. Traceability	Requirement-to-table-to-view-to-MVP matrix
6. Quality and feasibility	Normalisation, security, accessibility and risks

Executive summary

StudyCrew gives university project teams one private place to plan tasks, coordinate meetings and review factual contribution evidence. Its differentiator is not another task board: every meaningful collaboration action produces an immutable, inspectable event. The resulting dashboard supports earlier workload conversations without converting activity into an automated grade.

Design thesis Make work visible early enough to act, while keeping contribution evidence transparent, contextual and non-judgemental.

1. Product definition

1.1 Problem and opportunity

Small student teams often split coordination across chat, calendars and task lists. Status becomes stale, meeting attendance is hard to reconstruct and contribution concerns surface close to submission.

StudyCrew unifies planning and evidence so the team can see blocked work, upcoming commitments and activity history before delivery risk becomes critical.

1.2 Primary users and outcomes

Actor	Need	Successful outcome
Project member	Know what to do and where discussion belongs	Find assigned work, update status and collaborate without losing context
Project owner	Coordinate access and respond to delivery risk	Invite members, manage roles and identify blocked or overdue work
Whole team	Discuss contribution using shared facts	Reconcile activity totals with underlying events without a black-box score

1.3 Scope boundary

In scope: authenticated project collaboration, task workflow, comments, meetings, RSVP, activity evidence, in-app notifications, search/filter and evidence export. Out of scope: grading, sentiment analysis, peer scoring, LMS synchronisation, external email delivery, real-time chat and native mobile applications.

1.4 Assumptions

Assumption	Design response
A user may belong to multiple projects	Membership is a junction table; every project query is membership-scoped
One active owner must remain	A partial unique index permits one active owner and application logic prevents sole-owner removal
Evidence must survive ordinary edits	Source records use soft retention where needed and activity events are append-only
Users may be in different time zones	All instants use timestamp _{tz} /UTC and profiles store an IANA time zone

2. Functional requirements

The specification follows the assignment's IEEE 830-1998 Section 5.3.2 expectation. Each requirement is uniquely identified, uses mandatory shall language and defines inputs, processing, outputs, failure behaviour and a measurable acceptance condition. Sixteen of the eighteen functions are available only to authenticated users.

MVP rule The proposed MVP contains 15 requirements. FR-SEARCH-01 and FR-EXPORT-01 are designed extensions and are implemented only after the core quality gates pass.

FR-AUTH-01 · Register an account

Element	Specification
Shall statement	The system shall create a user account when a visitor submits a unique, syntactically valid email address, a display name of 2-80 characters and a password of at least 12 characters.
Input/trigger	visitor submits the registration form.
Processing	normalise the email to lowercase, reject an existing email, hash the password with the configured one-way password algorithm, create one user and one profile in a transaction, and start an authenticated session.
Success output	redirect to the empty project dashboard within 2 seconds under normal test load and display a registration confirmation.
Failure output	preserve non-secret form fields and identify each invalid field without revealing whether a different account is active.
Acceptance	valid input creates exactly one `users` row and one `profiles` row; duplicate email or invalid input creates neither row.
Trace	Tables: users; profiles Views: WF-01 MVP: Yes

FR-AUTH-02 · Sign in and sign out

Element	Specification
Shall statement	The system shall authenticate a registered user from an email and password and shall terminate that user's current session on sign-out.
Input/trigger	sign-in or sign-out submission.
Processing	compare the password using the configured hash verifier, rotate the session identifier after successful sign-in, and invalidate the server-side session on sign-out.
Success output	show the project dashboard after sign-in and the public sign-in view after sign-out.
Failure output	return one generic credential error and create no session.
Acceptance	an invalid password cannot access an authenticated route; a signed-out session cannot be reused to access one.
Trace	Tables: users Views: WF-01 MVP: Yes

FR-PROF-01 · Maintain a profile

Element	Specification
Shall statement	The system shall allow an authenticated user to view and update their display name, course, time zone,

Element	Specification
	biography and avatar URL.
Input/trigger	user submits profile fields.
Processing	validate lengths, permit only a recognised IANA time zone and update only the profile associated with the authenticated user.
Success output	show the saved profile values on refresh.
Failure output	identify invalid fields and leave stored values unchanged.
Acceptance	a user can update their own profile but cannot update another profile by changing a URL or request identifier.
Trace	Tables: profiles Views: WF-10 MVP: Yes

FR-PROJ-01 · Create a project

Element	Specification
Shall statement	The system shall allow an authenticated user to create a project with a name of 3-100 characters, optional description and due date.
Input/trigger	user submits the new-project form.
Processing	create the project and an 'owner' membership for the creator in one transaction.
Success output	open the new project overview showing the creator as owner.
Failure output	show validation errors and create no partial project.
Acceptance	successful creation produces exactly one project and one owner membership; the project is absent from non-members' dashboards.
Trace	Tables: projects; project_members Views: WF-02; WF-03 MVP: Yes

FR-PROJ-02 · Invite and join members

Element	Specification
Shall statement	The system shall allow a project owner to invite an email address and shall allow the matching authenticated user to accept or decline the invitation before its expiry.
Input/trigger	owner sends an invitation; invitee accepts or declines.
Processing	prevent duplicate pending invitations and existing-member invitations, expire invitations after 7 days, and create membership only on a valid acceptance.
Success output	show the pending invitation to both parties and, after acceptance, show the new member in the roster.
Failure output	reject expired, cancelled, duplicate or unauthorised use.
Acceptance	one accepted invitation creates one membership and cannot be accepted a second time.
Trace	Tables: project_invitations; project_members Views: WF-02; WF-06 MVP: Yes

FR-PROJ-03 · Administer project membership

Element	Specification
Shall statement	The system shall allow the project owner to change a member between `member` and `facilitator` roles or remove that member, but shall not allow removal or demotion of the sole owner.
Input/trigger	owner submits a role change or removal.
Processing	verify current ownership and target membership, then update or soft-remove the membership with an audit event.
Success output	refresh the roster and permission-dependent controls.
Failure output	return a permission or sole-owner constraint message.
Acceptance	a regular member receives HTTP 403 for the same operation; a removed member immediately loses project access.
Trace	Tables: project_members; activity_events Views: WF-06 MVP: Yes

FR-TASK-01 · Manage tasks

Element	Specification
Shall statement	The system shall allow a current project member to create, view, edit and archive a task with title, description, priority and optional due date.
Input/trigger	member submits a task create, update or archive action.
Processing	validate title length (3-120), priority (`low`, `medium`, `high`, `urgent`) and project membership; retain archived tasks for evidence.
Success output	update the project board without a full page reload and display the task's latest values.
Failure output	reject invalid or unauthorised changes without partial data.
Acceptance	valid create/edit/archive actions are visible after refresh; archived tasks are excluded from the default board and remain exportable.
Trace	Tables: tasks; activity_events Views: WF-03; WF-04; WF-05 MVP: Yes

FR-TASK-02 · Assign task responsibility

Element	Specification
Shall statement	The system shall allow a current project member to assign zero or more current project members to a task and to remove assignees.
Input/trigger	member submits the assignee selector.
Processing	reject users outside the task's project, avoid duplicate assignments and apply additions/removals atomically.
Success output	show the same assignees on the board and task detail view.
Failure output	identify an ineligible assignee and preserve prior state.

Element	Specification
Acceptance	assigning two eligible users creates two unique junction rows; resubmitting the same selection creates no duplicates.
Trace	Tables: task_assignees; project_members; notifications Views: WF-04; WF-05 MVP: Yes

FR-TASK-03 · Progress a task

Element	Specification
Shall statement	The system shall allow an assignee or project owner to move a task among `todo`, `in_progress`, `blocked` and `done` states.
Input/trigger	authorised user selects or drags a task to a state.
Processing	require a blocker note of 3-500 characters for `blocked`, set `completed_at` only for `done`, and record the actor and transition.
Success output	move the task to the selected board column and announce the change in an accessible live region.
Failure output	return the task to its prior state with a reason.
Acceptance	blocked without a note is rejected; moving to done sets one completion timestamp; reopening clears it.
Trace	Tables: tasks; activity_events Views: WF-04; WF-05 MVP: Yes

FR-COLL-01 · Discuss a task

Element	Specification
Shall statement	The system shall allow a current project member to add a plain- text comment of 1-2000 characters to a task and edit or delete their own comment.
Input/trigger	member submits a comment action.
Processing	escape active content, retain created/edited timestamps and permit project owners to moderate with an audit event.
Success output	append or update the comment thread in chronological order.
Failure output	reject blank, oversized or unauthorised modifications.
Acceptance	scripts render as text, not executable markup; a non-author, non-owner cannot modify the comment.
Trace	Tables: task_comments; activity_events; notifications Views: WF-05 MVP: Yes

FR-MEET-01 · Schedule a meeting

Element	Specification
Shall statement	The system shall allow a current project member to create, update or cancel a meeting with title, start time, end time, location/link and agenda.
Input/trigger	member submits the meeting form.

Element	Specification
Processing	require the end after the start, store UTC, retain organiser, and mark cancellation rather than deleting attendance evidence.
Success output	show the meeting in chronological project and calendar views.
Failure output	identify invalid times or fields and preserve stored data.
Acceptance	the same UTC meeting renders correctly in two test time zones; an end time equal to the start time is rejected.
Trace	Tables: meetings; activity_events; notifications Views: WF-03; WF-07 MVP: Yes

FR-MEET-02 · Respond to a meeting

Element	Specification
Shall statement	The system shall allow each current project member to set one response ('pending', 'accepted', 'declined') and an optional availability note for a non-cancelled meeting.
Input/trigger	member submits an RSVP.
Processing	upsert one response for the meeting/member pair and timestamp it.
Success output	update attendee counts and the member's displayed response.
Failure output	reject non-members and responses to cancelled meetings.
Acceptance	repeated changes update one row rather than creating duplicates; displayed counts equal the underlying response rows.
Trace	Tables: meeting_attendees; activity_events Views: WF-07 MVP: Yes

FR-CONTR-01 · Record contribution evidence

Element	Specification
Shall statement	The system shall append an immutable activity event when an authenticated member creates, updates or completes a task, comments, schedules a meeting, responds to a meeting or changes membership.
Input/trigger	successful completion of an audited action.
Processing	record actor, project, event type, relevant entity reference and UTC timestamp in the same transaction as the source action.
Success output	make the event available to the project activity feed.
Failure output	roll back the source action if its required event cannot be recorded; never expose secret or deleted comment content in metadata.
Acceptance	each audited source action creates exactly one matching event; application roles cannot update or delete event rows.
Trace	Tables: activity_events Views: WF-03; WF-07; WF-08 MVP: Yes

FR-CONTR-02 · View contribution insights

Element	Specification
Shall statement	The system shall allow a current project member to view per- member activity counts, completed tasks, comments and meeting participation for a selected project and date range of no more than 366 days.
Input/trigger	member selects date range and optional activity type.
Processing	aggregate immutable events and current task/attendance records; do not calculate a grade or qualitative score.
Success output	render summary totals, an accessible table and drill-down event list within 3 seconds under the seeded test dataset.
Failure output	identify invalid ranges and return no data for non-members.
Acceptance	dashboard totals reconcile exactly with a reference SQL query; zero-activity members remain visible with zero values.
Trace	Tables: activity_events; tasks; task_comments; meeting_attendees Views: WF-08; WF-11 MVP: Yes

FR-NOTIF-01 · Manage in-app notifications

Element	Specification
Shall statement	The system shall notify an authenticated user in-app when they are invited, assigned a task, mentioned in a comment, or affected by a meeting change, and shall allow each notification to be marked read.
Input/trigger	a qualifying source event or mark-read action.
Processing	create at most one notification per recipient/source event, suppress self-notification, and set `read_at` for the recipient only.
Success output	update unread count and link to the authorised source view.
Failure output	omit links to resources the recipient can no longer access.
Acceptance	duplicate event handling creates one notification; one user cannot mark another user's notification as read.
Trace	Tables: notifications; activity_events Views: WF-05; WF-07; WF-09 MVP: Yes

FR-SEARCH-01 · Search and filter project work

Element	Specification
Shall statement	The system shall allow a current project member to search task titles/descriptions and filter by status, priority, assignee and due-date state.
Input/trigger	member enters a query or changes a filter.
Processing	combine filters with logical AND, use case-insensitive text matching and return only tasks in the selected authorised project.
Success output	update results and report result count within 1 second after a 300 ms input debounce under the seeded test dataset.

Element	Specification
Failure output	show a clear empty state rather than stale results.
Acceptance	test fixtures return the documented result set for each filter and no cross-project task is returned.
Trace	Tables: tasks; task_assignees Views: WF-04 MVP: No

FR-EXPORT-01 · Export contribution evidence

Element	Specification
Shall statement	The system shall allow a current project member to request a PDF or CSV evidence export for an authorised project and date range.
Input/trigger	member selects format and valid date range.
Processing	create an export job, generate a snapshot containing project, range, generation time, member totals and underlying event rows, and expire the download after 24 hours.
Success output	show job status and provide an authorised download when ready.
Failure output	mark the job failed with a retry option and no partial file.
Acceptance	exported totals match FR-CONTR-02 for the same inputs; expired or non-member download requests are denied.
Trace	Tables: export_jobs; activity_events Views: WF-08; WF-11 MVP: No

FR-SEC-01 · Enforce project authorisation

Element	Specification
Shall statement	The system shall verify authentication and current project membership on every server-side read or mutation of project-scoped data.
Input/trigger	any request containing or resolving to a project identifier.
Processing	derive user identity only from the validated session and apply role checks before querying or mutating protected records.
Success output	allow the request only when the stated permission holds.
Failure output	return HTTP 401 when unauthenticated and HTTP 403 when authenticated but unauthorised, without returning protected record content.
Acceptance	an automated access-control matrix test covers every protected route with visitor, non-member, member and owner identities.
Trace	Tables: all project-scoped tables Views: WF-02 to WF-11 MVP: Yes

3. Data model

3.1 Modelling decisions

The physical model targets PostgreSQL 16. UUID keys are used for externally visible entities; bigint identity keys support high-volume immutable events and notifications. All instants use timestamptz. Enumerated states are represented by constrained varchar values so the allowed domain is explicit in the schema and ERD.

Required association	Implementation	Integrity evidence
One-to-one	users to profiles	profiles.user_id is both PK and FK; a profile cannot exist without one user
One-to-many	projects to tasks	tasks.project_id is NOT NULL; each task belongs to exactly one project
Many-to-many	users to projects	project_members has composite PK (project_id, user_id)
Many-to-many	users to tasks	task_assignees has composite PK (task_id, user_id)
Many-to-many	users to meetings	meeting_attendees has composite PK (meeting_id, user_id)

3.2 Table catalogue

Table	Purpose	Primary key	Principal relationships / constraints
users	Account identity and credential state	user_id	profiles, projects, memberships and authored records
profiles	One profile per account	user_id (PK/FK)	Exactly one-to-one with users
projects	Private team workspace	project_id	Created by user; owns memberships, tasks, meetings and events
project_members	User-project junction and role	project_id + user_id	Many-to-many users/projects; one active owner per project
project_invitations	Time-limited invitation lifecycle	invitation_id	Belongs to project; invited by user; unique pending email
tasks	Project work item and workflow state	task_id	Belongs to project; created by user; owns assignees/comments
task_assignees	Task-user assignment junction	task_id + user_id	Many-to-many tasks/users; no duplicate assignment
task_comments	Sanitised task discussion	comment_id	Belongs to task; authored/moderated by users
meetings	Scheduled or cancelled project meeting	meeting_id	Belongs to project; organised by user; owns attendance
meeting_attendees	One RSVP per meeting/member	meeting_id + user_id	Many-to-many meetings/users
activity_events	Append-only contribution evidence	event_id	Belongs to project/actor; optional single source target

Table	Purpose	Primary key	Principal relationships / constraints
notifications	Recipient-specific in-app alert	notification_id	Unique recipient/source-event pair
export_jobs	Asynchronous evidence export	export_job_id	Belongs to project/requester; bounded range and expiry

3.3 Normalisation

The schema satisfies third normal form. Repeating memberships, assignments and attendance are isolated in junction tables; non-key attributes describe only their table key; profile data is separated from credentials; derived contribution totals are calculated from events rather than stored; and notification/export lifecycle state is isolated from source collaboration records. JSONB is restricted to non-authoritative event metadata, while all reportable identities and relationships remain typed foreign keys.

Physical ERD 1/3 - Identity and project membership

PostgreSQL 16 types | Crow's Foot cardinality | PK, FK and UQ constraints shown

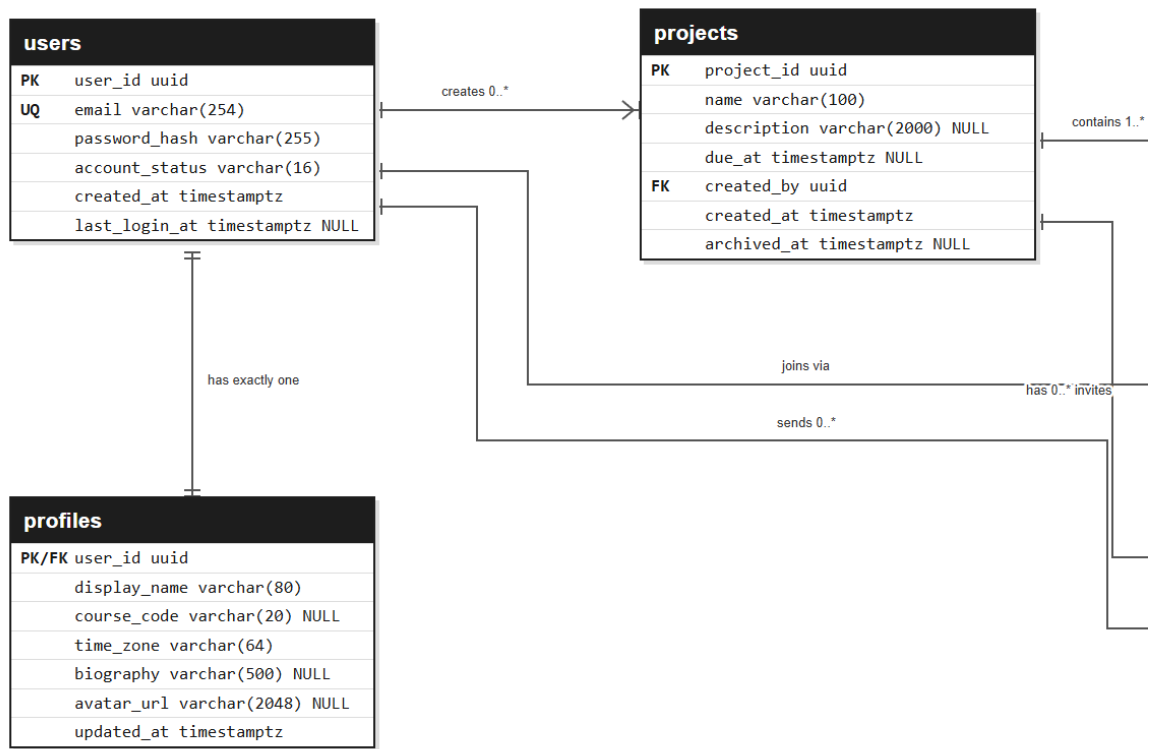


Figure 1A. Physical ERD 1/3 (left/centre view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

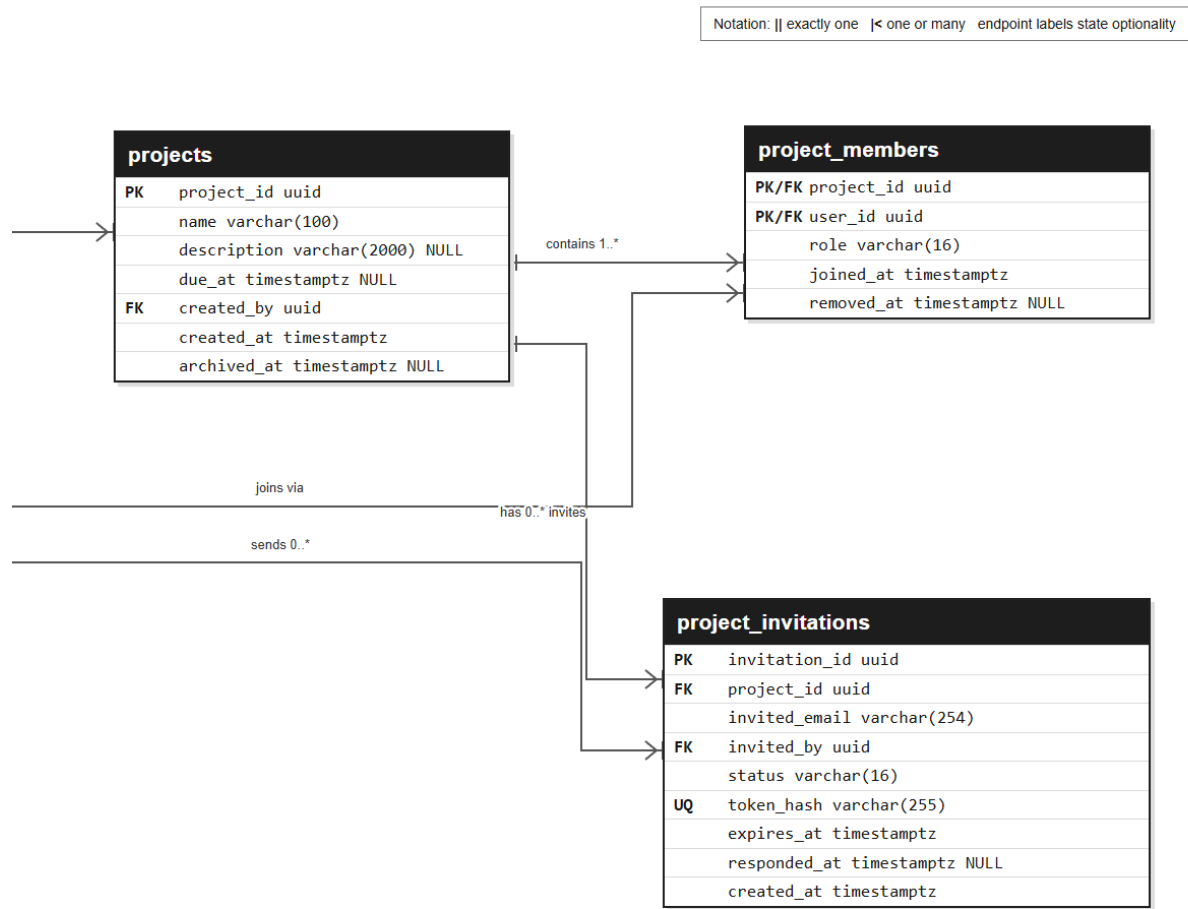


Figure 1B. Physical ERD 1/3 (centre/right view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

Physical ERD 2/3 - Tasks and collaboration

Anchor entities are repeated in light form to make every foreign-key relationship readable

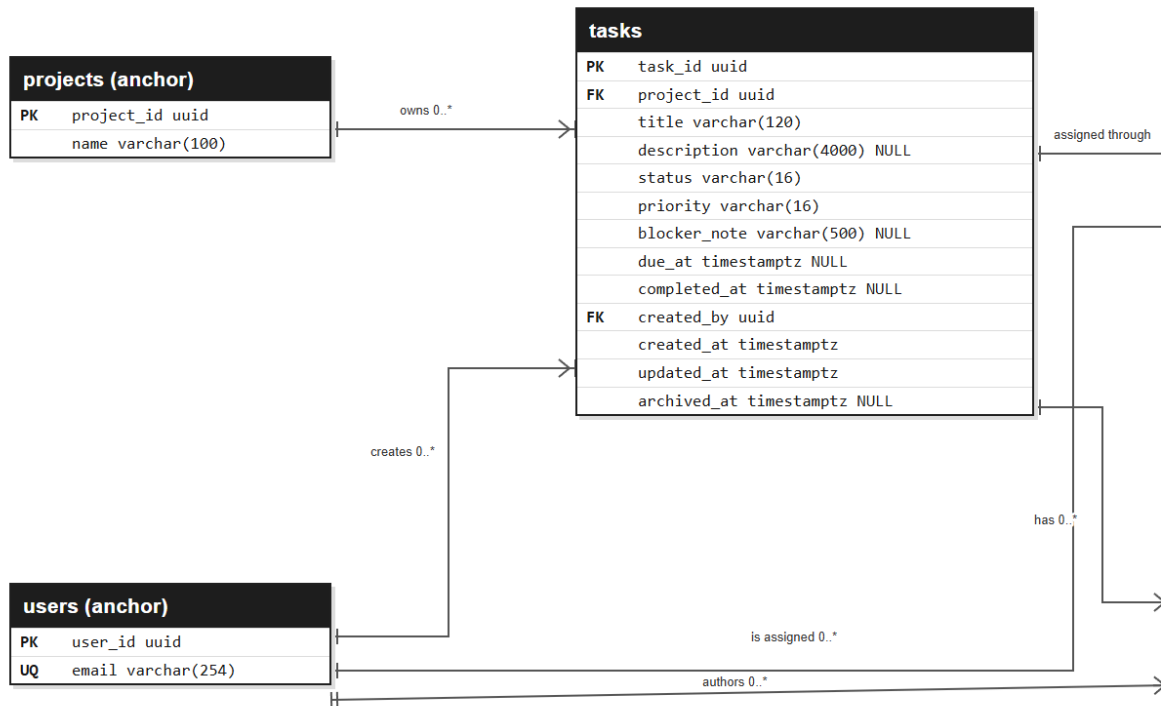


Figure 2A. Physical ERD 2/3 (left/centre view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

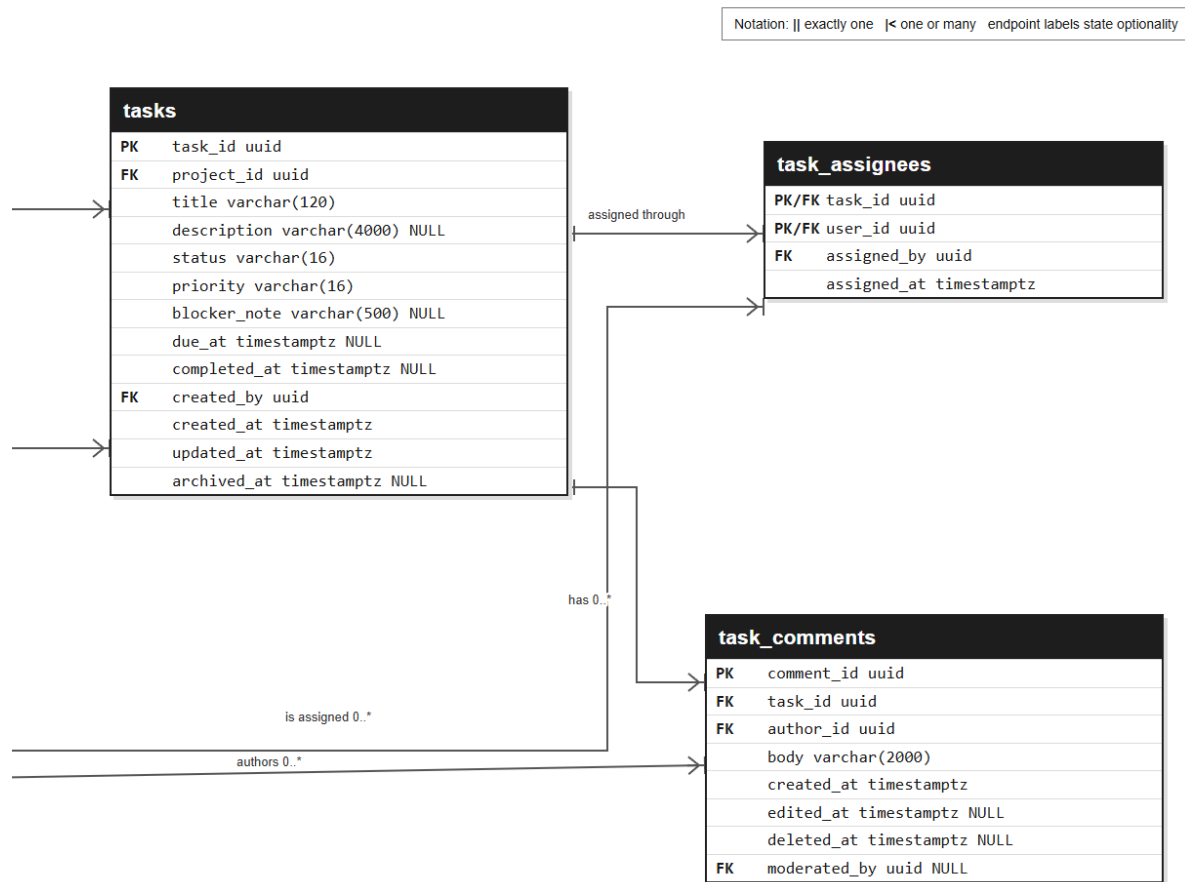


Figure 2B. Physical ERD 2/3 (centre/right view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

Physical ERD 3/3 - Meetings, evidence and exports

Contribution totals are derived from immutable events; no duplicated score or total is stored

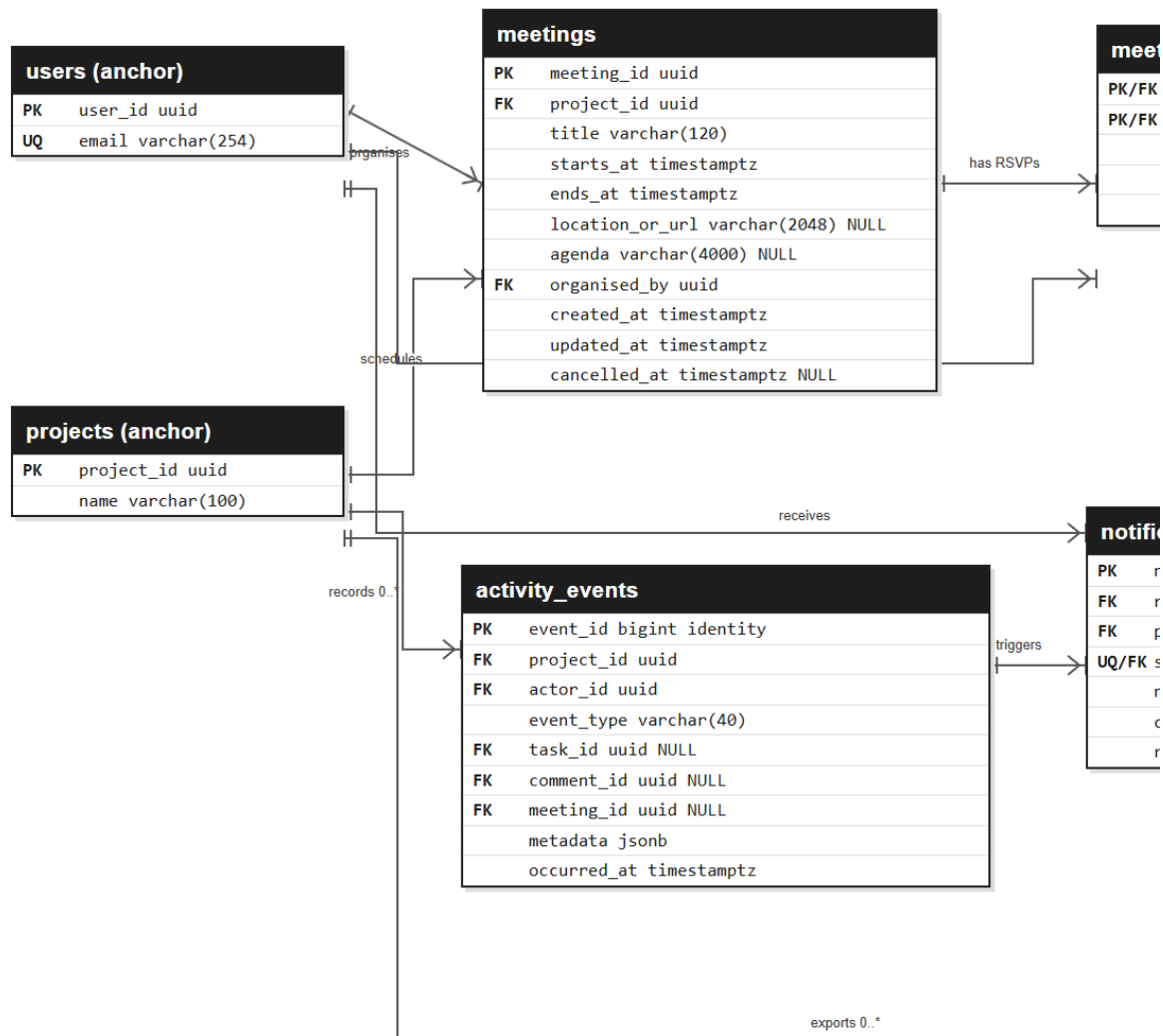


Figure 3A. Physical ERD 3/3 (left/centre view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

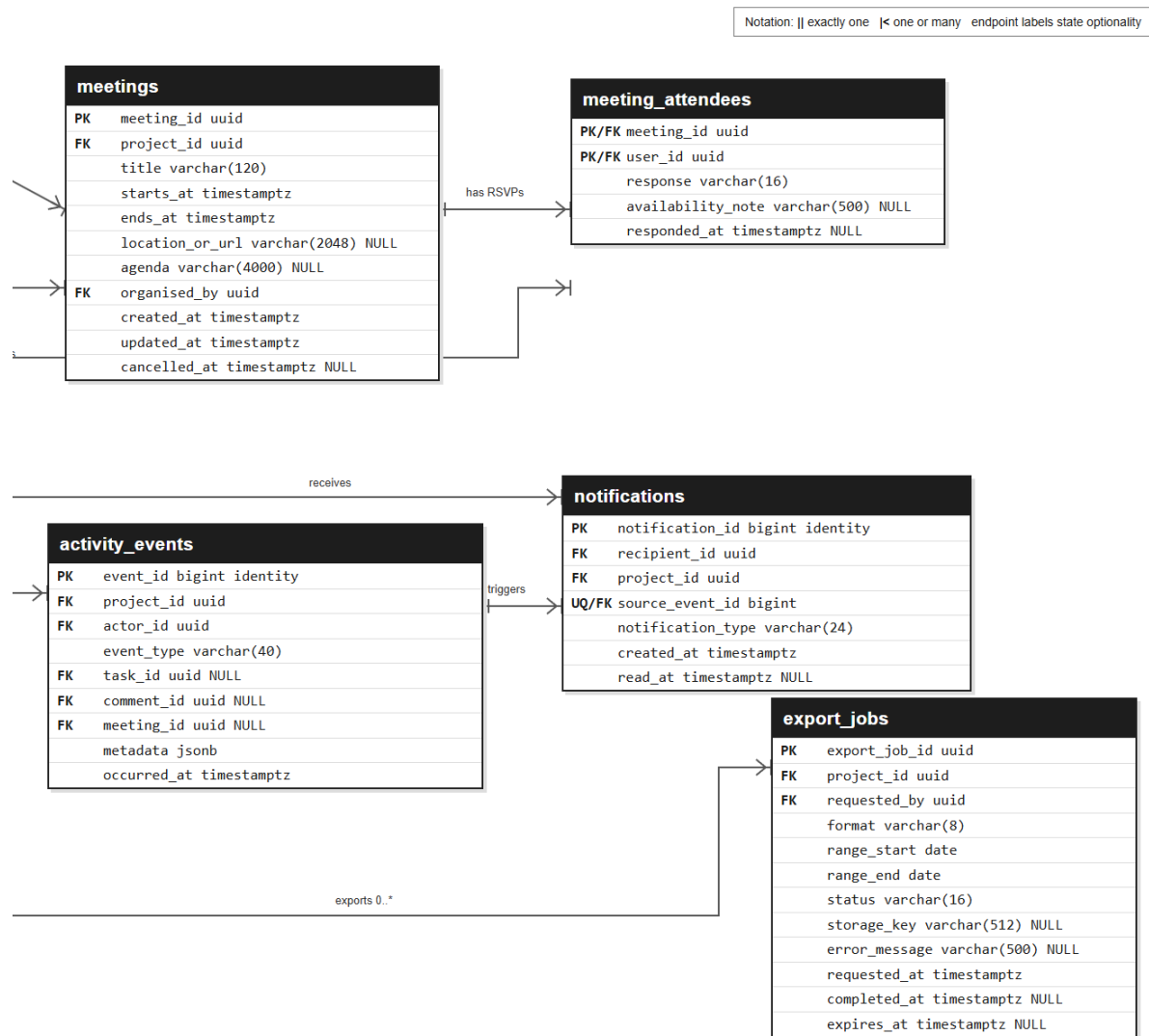


Figure 3B. Physical ERD 3/3 (centre/right view) in Crow's Foot notation. The deliberate overlap preserves relationship context; the complete landscape diagram is included in the repository.

3.4 Integrity and performance controls

Concern	Control
Duplicate identity	Lowercase email check plus unique users.email
Multiple active owners	Partial unique index on active owner membership
Duplicate invitations / notifications	Partial and composite unique indexes
Invalid workflow state	CHECK domains for task, meeting, invitation, export and notification state
Inconsistent completion	Task completion timestamp constraint tied to done status
Slow board and evidence queries	Composite indexes on project/status/priority and project/occurred_at
Evidence tampering	Append-only activity policy; application roles receive no UPDATE/DELETE grant

4. Wireframes

The wireframes intentionally omit colour and theming. They define information hierarchy, user controls, error/empty-state intent and cross-view flow. Requirement tags embedded in each frame make coverage directly inspectable.

WF-01 · Authentication

StudyCrew

Bring your group work into focus

Sign in to continue, or create an account.

UNIVERSITY EMAIL

student@uni.edu.au

PASSWORD

Sign in Create account

Registration panel

Display name, email, password (12+ characters), field-level validation.

Requirements demonstrated: FR-AUTH-01 FR-AUTH-02 Generic visitor entry and validation state.

Figure 4. WF-01 Authentication. Coverage: FR-AUTH-01, FR-AUTH-02.

WF-02 · Project dashboard

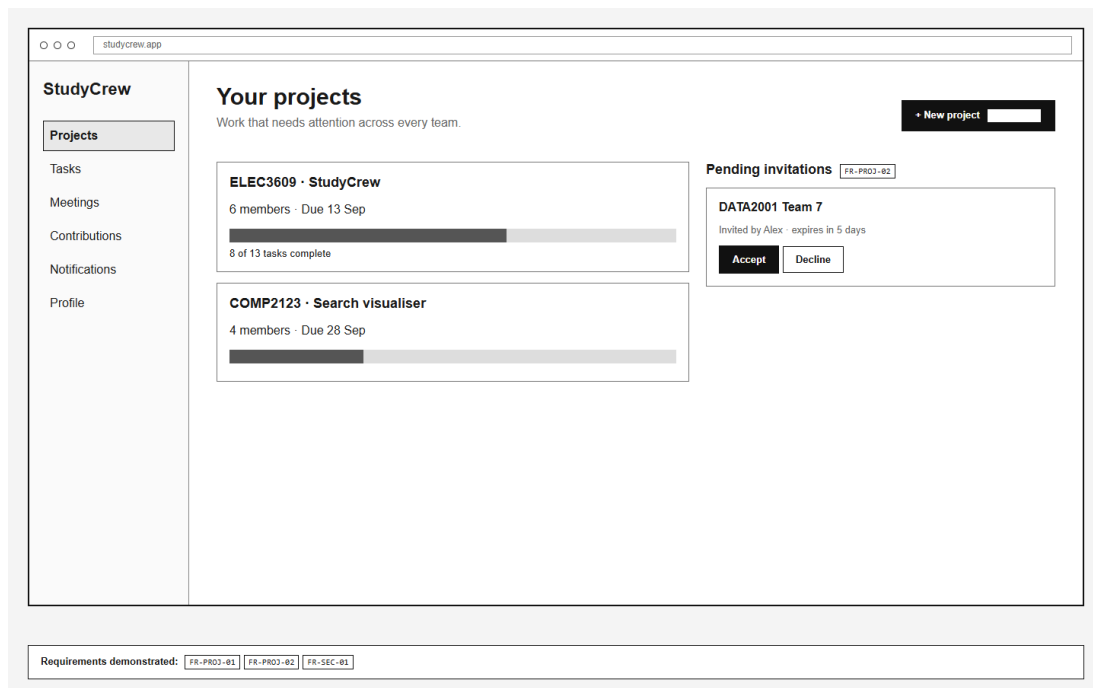


Figure 5. WF-02 Project dashboard. Coverage: FR-PROJ-01, FR-PROJ-02, FR-SEC-01.

WF-03 · Project overview

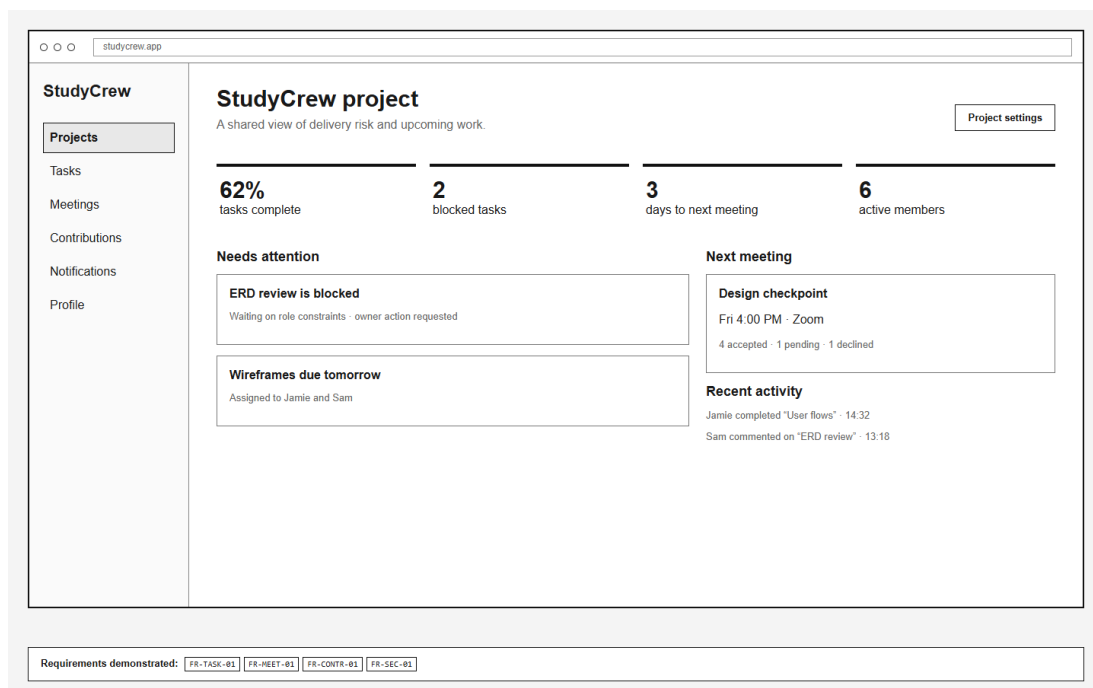


Figure 6. WF-03 Project overview. Coverage: FR-PROJ-01, FR-TASK-01, FR-MEET-01, FR-CONTR-01, FR-SEC-01.

WF-04 · Task board

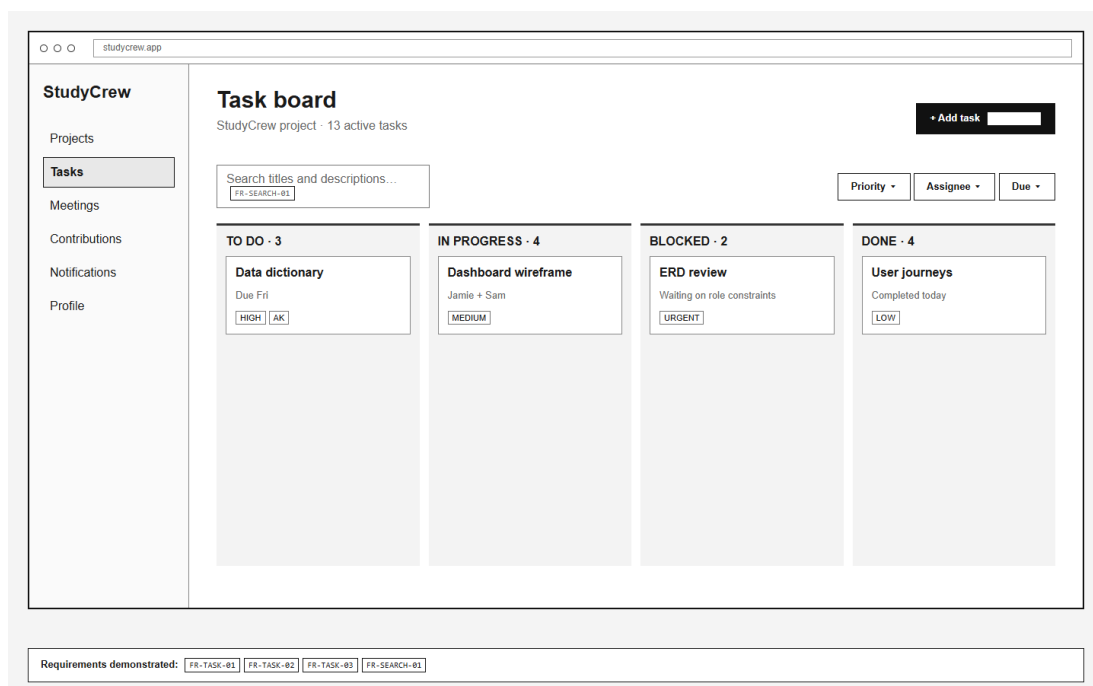


Figure 7. WF-04 Task board. Coverage: FR-TASK-01, FR-TASK-02, FR-TASK-03, FR-SEARCH-01, FR-SEC-01.

WF-05 · Task detail and discussion

The screenshot shows the 'StudyCrew' app interface. On the left is a sidebar with navigation links: Projects, Tasks (selected), Meetings, Contributions, Notifications, and Profile. The main content area is titled 'Tasks / ERD review' and 'ERD review'. It includes an 'Archive' button in the top right. The task details are as follows:

- DESCRIPTION:** Validate cardinalities, optionality, keys and PostgreSQL data types.
- STATUS:** Blocked (dropdown menu) with a tag 'FR-TASK-03'.
- PRIORITY:** Urgent (dropdown menu).
- BLOCKER NOTE (REQUIRED):** Waiting on owner-role constraint decision.
- ASSIGNEES:** AK - Jamie (dropdown menu) with a tag 'FR-TASK-02' and a '+ Add member' button.

On the right, the 'Discussion' section (tagged 'FR-COLL-01') shows two comments:

- Jamie** · 13:18: Should the owner be represented by membership role?
- AK** · 13:24 · edited: Yes. A partial unique index guarantees one active owner.

Below the comments is a text input field 'Write a comment... @mention a teammate' and a 'Comment' button.

At the bottom, a 'Requirements demonstrated:' section lists tags: FR-TASK-01, FR-TASK-02, FR-TASK-03, FR-COLL-01, FR-NOTIF-01, and FR-SEC-01.

Figure 8. WF-05 Task detail and discussion. Coverage: FR-TASK-01, FR-TASK-02, FR-TASK-03, FR-COLL-01, FR-NOTIF-01, FR-SEC-01.

WF-06 · Members and invitations

The screenshot shows the 'StudyCrew' app interface for 'Members & invitations'. The sidebar on the left has 'Projects' selected. The main content area is titled 'Members & invitations' and includes a note: 'Owner-only controls are hidden from regular members.' and an 'Invite member' button.

Member	Role	Joined	Actions
Alex Kim alex@uni.edu.au	Owner	20 Aug	Sole owner - protected
Jamie Lee	Facilitator (dropdown menu)	20 Aug	Remove
Sam Patel	Member (dropdown menu)	21 Aug	Remove

Below the table, the 'Pending' section shows an invitation: 'newmember@uni.edu.au · expires 27 Aug' with a 'Cancel invite' button.

At the bottom, a 'Requirements demonstrated:' section lists tags: FR-PROJ-02, FR-PROJ-03, and FR-SEC-01.

Figure 9. WF-06 Members and invitations. Coverage: FR-PROJ-02, FR-PROJ-03, FR-SEC-01.

WF-07 · Meetings and RSVP

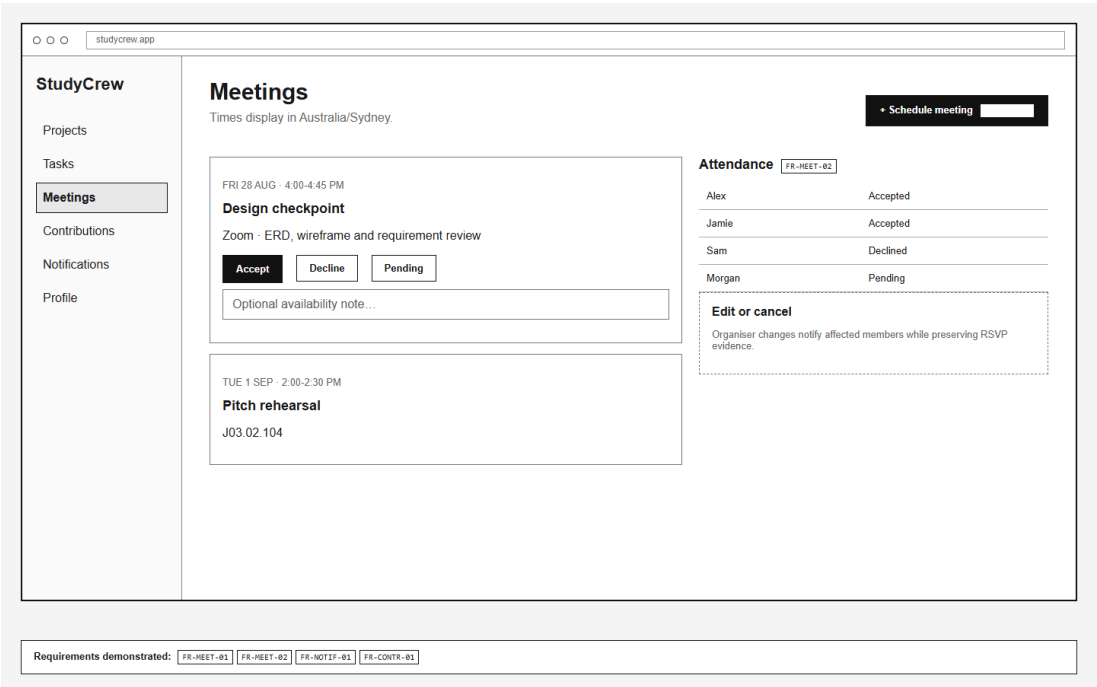


Figure 10. WF-07 Meetings and RSVP. Coverage: FR-MEET-01, FR-MEET-02, FR-CONTR-01, FR-NOTIF-01, FR-SEC-01.

WF-08 · Contribution evidence

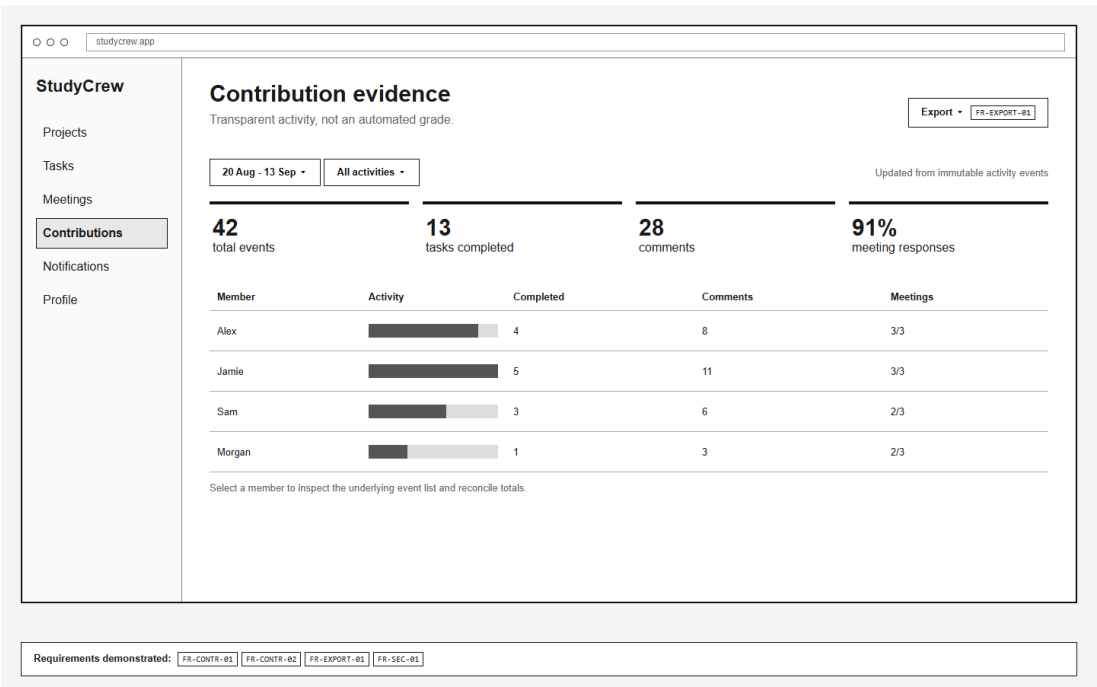


Figure 11. WF-08 Contribution evidence. Coverage: FR-CONTR-01, FR-CONTR-02, FR-EXPORT-01, FR-SEC-01.

WF-09 · Notifications

StudyCrew

Projects

Tasks

Meetings

Contributions

Notifications

Profile

Notifications

3 unread · newest first Mark all read

Jamie assigned you "Dashboard wireframe" UNREAD

StudyCrew project · 5 minutes ago

Open task

Design checkpoint changed to 4:00 PM UNREAD

StudyCrew project · 18 minutes ago

Review meeting

You were invited to DATA2001 Team 7 UNREAD

2 hours ago · expires in 5 days

Review invitation

Sam mentioned you in "ERD review"

Yesterday · read

Requirements demonstrated: FR-NOTIF-01 FR-SEC-01 Links are resolved through current authorisation; inaccessible sources are not exposed.

Figure 12. WF-09 Notifications. Coverage: FR-NOTIF-01, FR-SEC-01.

WF-10 · Profile

StudyCrew

Projects

Tasks

Meetings

Contributions

Notifications

Profile

Your profile

Used for team identity and meeting time display. Save changes

Avatar URL

AK

DISPLAY NAME

COURSE

TIME ZONE

BIOGRAPHY

Privacy boundary

Only current project teammates can see project-linked profile information.

Account email and password controls are not exposed on project pages.

Requirements demonstrated: FR-PROF-01 FR-SEC-01

Figure 13. WF-10 Profile. Coverage: FR-PROF-01, FR-SEC-01.

WF-11 · Evidence export

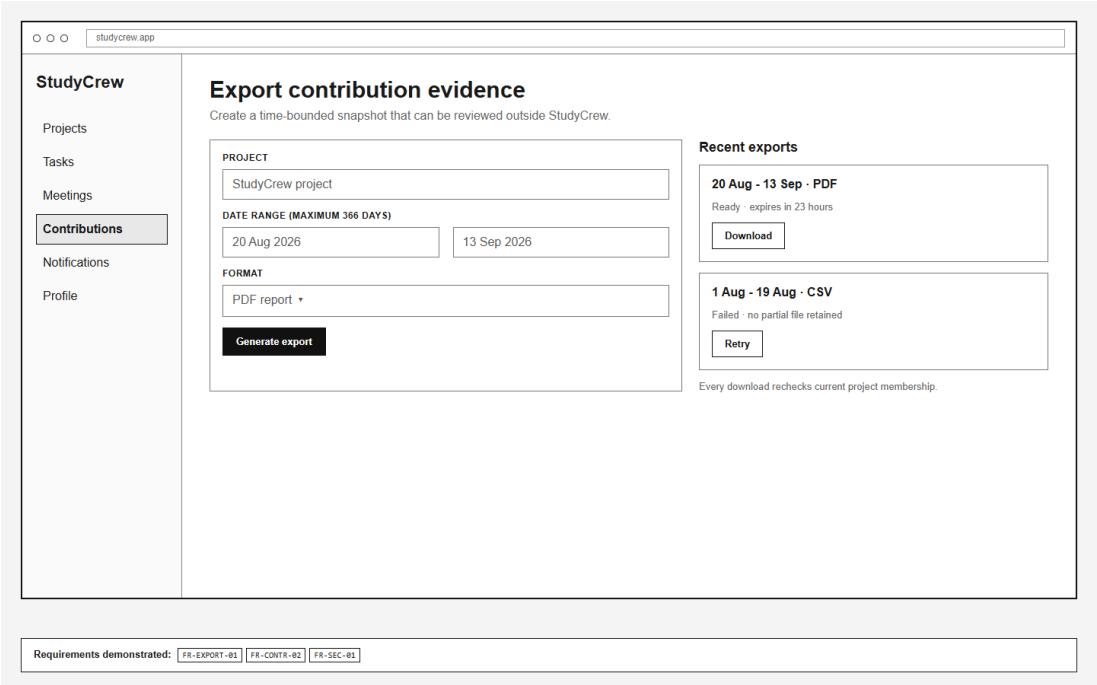


Figure 14. WF-11 Evidence export. Coverage: FR-CONTR-02, FR-EXPORT-01, FR-SEC-01.

4.1 End-to-end user flows

Flow	Views	Completion evidence
Start a team	WF-01 → WF-02 → WF-06 → WF-03	Owner account, project and accepted membership exist
Deliver a task	WF-04 → WF-05 → WF-04	Task assigned, discussed and moved to done with event record
Coordinate a meeting	WF-07 → WF-09	Valid UTC meeting, one RSVP per member and change notification
Review contribution	WF-08 → event drill-down → WF-11	Totals reconcile to immutable events and authorised export

5. Traceability matrix

Requirement	Primary tables	Wireframes	MVP
FR-AUTH-01	users; profiles	WF-01	Yes
FR-AUTH-02	users	WF-01	Yes
FR-PROF-01	profiles	WF-10	Yes
FR-PROJ-01	projects; project_members	WF-02; WF-03	Yes

Requirement	Primary tables	Wireframes	MVP
FR-PROJ-02	project_invitations; project_members	WF-02; WF-06	Yes
FR-PROJ-03	project_members; activity_events	WF-06	Yes
FR-TASK-01	tasks; activity_events	WF-03; WF-04; WF-05	Yes
FR-TASK-02	task_assignees; project_members; notifications	WF-04; WF-05	Yes
FR-TASK-03	tasks; activity_events	WF-04; WF-05	Yes
FR-COLL-01	task_comments; activity_events; notifications	WF-05	Yes
FR-MEET-01	meetings; activity_events; notifications	WF-03; WF-07	Yes
FR-MEET-02	meeting_attendees; activity_events	WF-07	Yes
FR-CONTR-01	activity_events	WF-03; WF-07; WF-08	Yes
FR-CONTR-02	activity_events; tasks; task_comments; meeting_attendees	WF-08; WF-11	Yes
FR-NOTIF-01	notifications; activity_events	WF-05; WF-07; WF-09	Yes
FR-SEARCH-01	tasks; task_assignees	WF-04	No
FR-EXPORT-01	export_jobs; activity_events	WF-08; WF-11	No
FR-SEC-01	all project-scoped tables	WF-02 to WF-11	Yes

Coverage audit: 18/18 requirements map to at least one table and one wireframe. All 13 tables appear in the ERDs. The 15 MVP requirements map to concrete acceptance conditions; the two post-MVP features remain designed and traceable.

6. Quality, feasibility and risk

6.1 Security and privacy

Server-side authorisation derives identity from the session and checks current membership before every project-scoped query or mutation. Password hashes, invitation token hashes and account emails never enter activity metadata or exports. Soft removal immediately revokes project access. Exports expire after 24 hours and each download rechecks membership.

6.2 Accessibility and responsive behaviour

Interactive controls use visible labels and keyboard focus; task status changes are announced in an accessible live region; charts have equivalent tables; errors identify fields in text; and colour is never the sole state indicator. At narrow widths, the sidebar becomes a menu, board columns scroll horizontally with labelled regions and forms collapse to one column without changing function.

6.3 Feasibility

Layer	Proposed dependency	Reason
Client	React + accessible component primitives	Responsive views and predictable state-driven interactions
API	Node.js + Express	Small-team familiarity and testable REST boundary
Data	PostgreSQL + Prisma	Relational integrity, transactions and explicit migrations
Testing	Vitest + Playwright	Unit, integration, access-control and end-to-end journeys
Delivery	GitHub Actions + Docker	Repeatable checks and environment parity

6.4 Principal risks

Risk	Likelihood / impact	Mitigation and trigger
Activity counts interpreted as grades	Medium / High	No score; show definitions and event drill-down; revise language if users infer quality
Scope expansion	High / Medium	Freeze approved MVP; search/export remain post-MVP until quality gates pass
Authorisation leak	Low / High	Central policy middleware and route-by-role matrix in CI
Late integration	Medium / High	Vertical slices from week 7; seeded journeys pass at each milestone

References

- [1] IEEE Computer Society, IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, 1998. Superseded standard; used because the assignment explicitly requires Section 5.3.2.
- [2] ELEC3609/9609 Group Project, Assignment 1: System & Database Design, 2026.

Schema source: database/schema.sql. Requirement source: docs/requirements.md. Traceability source: docs/traceability.csv.